

# KARTIK SIROHI

kartik.orion.dev@gmail.com | github.com/sirohikartik | +91-7906443460 | NIT Jalandhar, Punjab, India

## INTERESTS

Performance Engineering - memory-efficient systems, paging and storage-aware indexing, SIMD/vectorization, cache-aware computation, low-level ML runtime optimization, and high-performance biological sequence search.

## EDUCATION

**NIT Jalandhar**  
*B.Tech, Information Technology | CGPA: 9.5/10.0*

Punjab, India  
2024 – Present

## EXPERIENCE

**Project Lead — EPSE (Efficient Protein Search Engine)**

Jun 2026 – Present

*Open Source | [github.com/sirohikartik/epse](https://github.com/sirohikartik/epse)*

*Remote*

- Building a **memory-efficient protein sequence search engine in C**, focused on **paging-based indexing** to minimize RAM usage during large-scale biological sequence retrieval.

**AI Research Intern**

Jan 2026 – Mar 2026

*University of Galway*

*Remote*

- Co-authored **DICE-RoBERTa** (accepted, ComSys 2026) — a compute-efficient video classification model achieving **85.0% accuracy on UCF101 at only 9.8 GFLOPs/video**, a 5–10× reduction vs. baseline transformers.
- Second paper on neural networks in computational psychiatry currently **under review at IEEE Access**.

**Freelance AI / Backend Engineer**

Aug 2025 – Dec 2025

*Independent*

*Remote*

- Built **FasTest**, an AI-powered API testing platform with automated test generation and concurrent execution.
- Designed a context-aware LLM pipeline producing structured JSON payloads for edge-case coverage.

## PROJECTS

**tinyGPT** — C++ Transformer Inference Engine | *C++*, *SIMD (AVX/NEON)*

2026

- Built a **30M parameter decoder-only transformer** from scratch with a custom C++ inference engine.
- Implemented **SIMD-optimized matrix multiplication** (AVX/NEON) and **cache-aware tiled kernels** for constrained environments.
- Engineered localized execution to achieve a **96.9ms Time-to-First-Token (TTFT)** for prompt processing and a generation throughput of **1.88 tokens/sec (537ms/token)** on a baseline consumer Apple M1 Air (8GB RAM).

**Mini-Torch** — Neural Network Framework | *Python*

2026

- Built a custom tensor engine with **reverse-mode automatic differentiation** and dynamic computation graph.
- Implemented backpropagation from scratch; SGD and Adam optimizers.

## OPEN SOURCE CONTRIBUTIONS

**ggml-org/llama.cpp** — PR #22209 (Merged)

2026

- Vectorized ggml\_vec\_dot\_q4\_1.q8\_1** — rewrote hot-path quantized dot product using SIMD intrinsics; directly improves LLM inference throughput.

**pytorch/pytorch** — PR #172200 (Merged)

2026

- Fixed **NaN propagation bug in tensor clamp operation**; navigated pytorch's large codebase and strict review process independently.

**alexandria3k** — PR #63 (Merged)

2026

- Reduced lookup complexity from **O(n)** to **O(1)** via algorithmic restructuring.

**scipy/scipy** — PR #23129 (Merged)

2025

- Added documentation examples to Gauss-Jacobi quadrature API.

**Tracer-Cloud/opensre** — PR #852, PR #917, PR #1059 (Merged)

2026

- Added unit tests for data validation and Grafana Tempo mixin in opensre.

## TECHNICAL SKILLS

**Performance / Systems:** SIMD (AVX, NEON), Cache-aware algorithms, Memory optimization, Low-level C++

**AI / ML:** Transformers, LLMs, CNNs, RAG, Autograd, FAISS, Quantization

**Languages:** C++, Python, JavaScript

**Backend:** FastAPI, REST APIs, Async Systems, PostgreSQL

**Tools:** Git, Linux, Postman, Vercel

**Core CS:** Data Structures & Algorithms, Operating Systems, OOP, DBMS, Computer System Architecture

## ACHIEVEMENTS

- Ranked **1365 / 34,000+** participants in LeetCode Biweekly Contest.
- Secured **3rd place** at HackMol 2025, NIT Jalandhar.